

Nicola Bottura, PhD

WINDOWS LOW-LEVEL C++ ENGINEER · ENDPOINT SECURITY & MALWARE ANALYSIS

Rome, Italy

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Summary

Current R&D at Prisma s.r.l., working on IoT security and Malware analysis and doing scientific research with the Sap4Sec research group at Sapienza University of Rome where I recently (Jan. 2026) got my PhD. 5+ years of experience in malware analysis and reverse engineering, binary instrumentation and fuzzing techniques. Published in relevant security conferences (IEEE EuroS&P, DIMVA) and granted a patent (Italy, Europe, and US). My background helped me in gaining interest and expertise also in vulnerability finding and offensive tradecrafting.

Work Experience

Prisma s.r.l.

Rome, Italy

RESEARCH & DEVELOPMENT

Nov. 2025 - Present

- Designing and implementing PFuzzer as a sandbox-integrated malware analysis platform for commercial deployment (Windows, C++), focusing on robustness and low-overhead dynamic instrumentation.
- Automating reverse engineering workflows for Windows malware using program analysis techniques (coverage-guided exploration, behavioral telemetry, anti-analysis detection).

CINI Consorzio Interuniversitario Nazionale per l'Informatica

Rome, Italy

RESEARCH FELLOW

March 2022 - May 2022

- Research on novel methods to automate reverse engineering efforts in malware with program analysis.

Education

Sapienza University of Rome

Rome, Italy

PHD IN ENGINEERING IN COMPUTER SCIENCE

May. 2022 - Jan. 2026

- **Thesis title:** Engaging with Malware: Coverage-guided Fuzzing and Dynamic Introspection to Analyze Environment Sensitive and Obfuscated Windows Malware.
- Visited for 3 months University of Twente in Enschede (NL) and collaborated with Prof. Andrea Continella

Sapienza University of Rome

Rome, Italy

M.Sc. IN CYBERSECURITY

Sep. 2019 - Jan. 2022

- **Thesis title:** Revisiting API Interposition Techniques.

Unimore (University of Modena and Reggio Emilia)

Modena, Italy

B.Sc. IN COMPUTER SCIENCE

Sep. 2015 - Jul. 2019

- **Thesis title:** Design and Development of a Mechanism to Reduce the Guaranteed Latency of the BFQ Scheduler.
- Contributed with a [commit] for Linux with my patch to BFQ

Technical Skills

Programming Languages

C, modern C++, Python, x86-64 Assembly, PowerShell, Bash

Windows Internals

Low-level Windows development, Win32 API, Native API, Memory management, COM, Registry

Reverse Engineering

IDA Pro, DBI, Binary Ninja, Ghidra, WinDbg, Static/Dynamic analysis

Malware Analysis

Behavioral analysis, anti-analysis techniques, Obfuscation, Malware PoC development, EDR, anti-tampering

Vulnerability Research

Fuzzing, Harness development, Dynamic Program Analysis, Memory safety

Tooling

DynamoRIO, Intel Pin, AFL, libDFT, YARA, Git

Publications

2025	PDF	Pfuzzer: Practical, Sound, and Effective Multi-path Analysis of Environment-sensitive Malware with Coverage-guided Fuzzing N. Bottura, D. C. D'Elia, L. Querzoni <i>IEEE EuroS&P</i>
2025	PDF	All Right Then, (Don't) Keep Your Secrets: Exposing API Hashing in Malware N. Bottura, G. Di Pietro, Y. Yamada, D. C. D'Elia, L. Querzoni <i>DIMVA</i>

Projects

PFuzzer

DynamoRIO-based coverage-guided fuzzer (17k+ LoC in C/C++).
Published; granted a patent; ongoing work toward industrial applicability (currently owned by Prisma s.r.l., code cannot be shared).

Unprotect Project

Contributed with four novel anti-analysis techniques found in malware - link.
Unprotect is an open source project that gathers knowledge on anti-analysis techniques in malware.

Offensive Tradecrafting

Maintain a public repository with PoC prototypes and write-ups supporting analysis and detection tooling - link.
Includes malware techniques found during my research and other techniques I found interesting in re-developing.